

Fluid Mechanics

LUO Min & BAI Qiang

Institute of Port, Coastal & Offshore Engineering

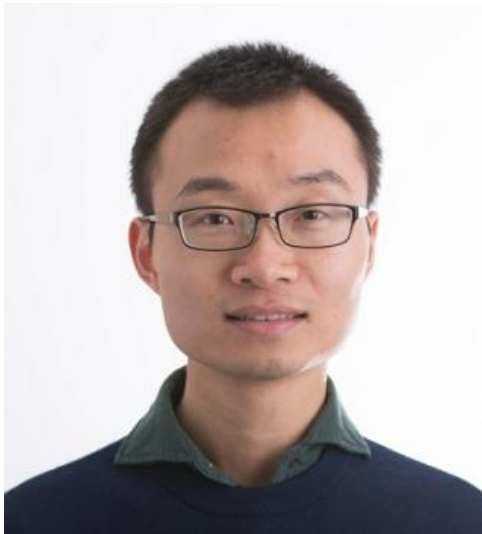
E-mails: min.luo@zju.edu.cn, qiangbai@zju.edu.cn

Course Objective

A core course for undergraduate students of Ocean Engineering/Technology

- Understanding of the **fundamental concepts** of fluid and **basic theories** of fluid motion, as well as the ability to **use fluid mechanics to analyse and solve fluid-related problems**
- **The foundation for learning more advanced courses**, and prepares students for engaging with professional and technical work in various disciplines in ocean engineering

Delivery and Assessment Methods



Course delivery

- Class teaching/discussing (in English) in紫金港西 1-505;
- Classes 3-5 on Friday
- Homework (written in English)

Assessment

- A closed-book final exam (60%, questions exactly the same as those of other classes, but in English)
- Homework & class engagement (40%)

Course Syllabus

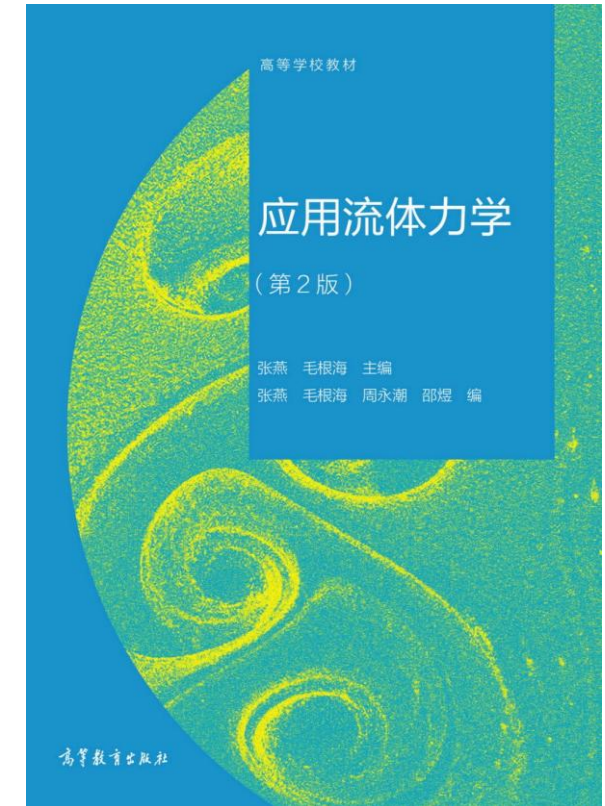
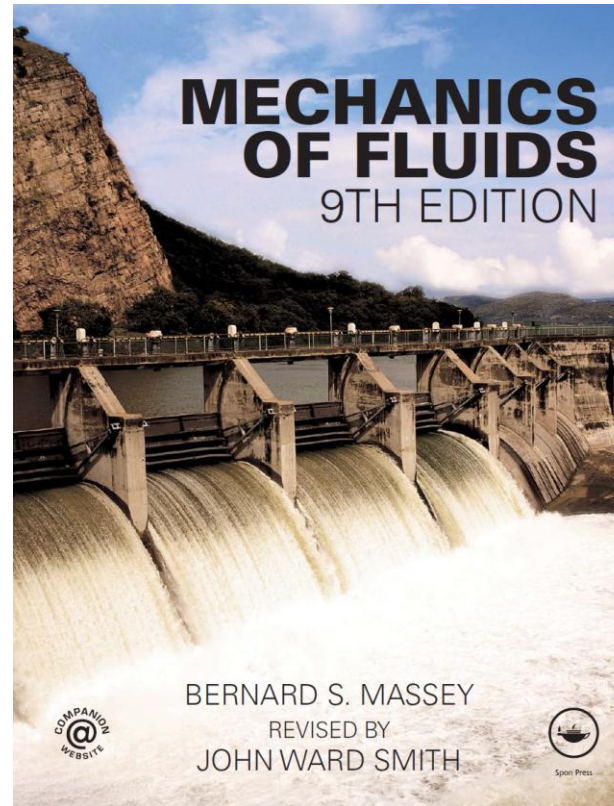
Week	Topics	Teaching hour (48)	Lecturer
1	Introduction and Basic Concepts of Fluid Mechanics: History of Fluid Mechanics, Definition, Inertia, and Compressibility of Fluid	3 hours	Bai
2	Basic Concepts of Fluid Mechanics: Viscosity, Surface tension, Vapor pressure of Fluid, Forces acting on Fluid	3 hours	Bai
3	Hydrostatics: Stress equilibrium, Pressure profile of static fluid, Hydrostatic pressure	3 hours	Bai
4	Hydrostatics: Buoyancy and floating body stability; Fluid Dynamics; Fundamentals of Fluid Motion: How to describe fluid motion	3 hours	Bai
5	Hydrodynamics: Basic concepts of fluid flow; Motion of a fluid parcel	3 hours	Bai
6	Hydrodynamics: Basic equations of fluid motion; Integration of Euler equations	3 hours	Luo
7	Hydrodynamics: Potential flow; Summary of Part 1	3 hours	Bai
8	Steady Total Flow: Basic equations & Bernoulli equation	3 hours	Luo

Course Syllabus

Week	Topics	Teaching hour (48)	Lecturer
9	5.1 holiday, no class		
10	Steady Total Flow: Bernoulli equation	3 hours	Luo
11	Steady Total Flow: Bernoulli equation Dimensional Analysis and Similarity Principle	1 hour 2 hours	Luo
12	Dimensional Analysis and Similarity Principle Flow resistance and Head Loss: Fundamentals	2 hour 1 hour	Luo
13	Flow resistance and Head Loss: Fundamentals Flow resistance and Head Loss: Turbulence	2 hours 1 hour	Luo
14	Flow resistance and Head Loss: Turbulence	3 hours	Luo
15	Flow resistance and Head Loss: Local head loss Flow resistance and Head Loss: Cd Course Summary and Review	2.5 hours 0.5 hour	Luo
16	Dragon Boat Festival, no class		

Recommended Textbooks

- Mechanics of Fluids (9th edition), B. S. Massey, revised by J. Ward-Smith, Spon Spress, 2012.
- 《应用流体力学》，毛根海主编/邵卫云，张燕编，2006年11月，高等教育出版社出版，（“十一五”国家级规划教材）



Sincere Advices

- All the teaching, discussions, and the final exam in the class will be in English. So, it will be **more difficult (at least in the beginning) than the classes in Chinese.**
- If you want to challenge yourself by learning and discussing in English, you are very welcome to join our class. We, both teachers, will do our best to explain the fundamental concepts of Fluid Mechanics clearly and systematically.
- If you just randomly selected our class and are not that keen on learning in English, please think about the class more carefully. To learn the course, knowledge is more important than the language.
- **If you stay in the class, please enjoy it, but do not complain about the teaching in English later.**